

Joint space control in SMC

Marko Guberina

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0.1 Joint-space control

There are just 2 controllers. A P-controller with feedback on positions for point to point motion and a P-controller with feedback on positions and feedforward velocity for trajectory following. Since they get the job done for the rare cases they are needed in, they haven't been developed further. There is a PI controller for point to point motion as well, but it's never been used for anything.

moveJP

1. $e = q_{\text{desired}} \ominus q$
2. $v_{cmd} = K \cdot \|e\| \cdot \frac{1}{freq}$

Note that the controller below is implemented in a very ugly way in code.

followKinematicJointTrajP

1. $q_{\text{desired}} = \text{interpolate}(q_{\text{reference}}(t))$
2. $q_{\text{error}} = q_{\text{desired}} - q$
3. $v_{\text{desired}} = \text{interpolate}(v_{\text{reference}}(t))$
4. $v_{cmd} = v_{\text{desired}} + K_p \cdot q_{\text{error}}$